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Experiment 2 – To Perform Data Preprocessing Using Python

Aim

To perform data preprocessing on a synthetic Video Streaming Platform dataset using Python by handling missing values, removing duplicate records, encoding categorical data, and scaling numerical features.

Objectives

- Load the streaming platform dataset using Python.
- Identify and handle missing values.
- Remove duplicate records.
- Convert categorical data into numerical form.
- Normalize/standardize numerical data.
- Prepare clean data for further analysis or machine learning.

Software/Tools Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Pandas
- NumPy
- Scikit-learn

Theory

Data preprocessing is the process of converting raw, incomplete, or inconsistent data into a clean and suitable format for analysis and machine learning.

Common preprocessing steps are:

1. Data Cleaning – Handling missing and incorrect values.
2. Removing Duplicates – Eliminating repeated records.
3. Encoding – Converting categorical values into numerical values.
4. Feature Scaling – Bringing numerical features to a similar scale.
5. Data Transformation – Converting data into a suitable format.

Dataset

A synthetic dataset of 515 records was generated to represent streaming activity on the Video Streaming Platform, mirroring the Fact_Streams schema designed in Experiment 1. The dataset was intentionally seeded with missing values, duplicate records, and inconsistent categorical text to give every preprocessing step meaningful data to act on.

Columns: user_id, age, subscription_tier, device_type, watch_duration_min, completion_pct, genre, country, monthly_spend

Python Program

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder, StandardScaler

# 1. Load Dataset
df = pd.read_csv('streaming_platform_dataset.csv')
print(f"Original Shape: {df.shape}")

# 2. Remove Duplicate Records
duplicates = df.duplicated().sum()
print(f"Duplicates Found: {duplicates}")
df = df.drop_duplicates()
print(f"Shape After Dropping Duplicates: {df.shape}")

# 3. Handle Missing Values
# Numerical imputation using mean
num_cols = ['age', 'watch_duration_min', 'completion_pct']
for col in num_cols:
    df[col] = df[col].fillna(df[col].mean())

# Categorical imputation using mode (most frequent value)
df['country'] = df['country'].fillna(df['country'].mode()[0])

# 4. Encode Categorical Data
cat_cols = ['subscription_tier', 'device_type', 'genre', 'country']
label_encoder = LabelEncoder()
for col in cat_cols:
    df[col] = label_encoder.fit_transform(df[col])

# 5. Feature Scaling (Standardization: Mean=0, Std=1)
scale_cols = ['age', 'watch_duration_min', 'completion_pct', 'monthly_spend']
scaler = StandardScaler()
df[scale_cols] = scaler.fit_transform(df[scale_cols])

# Display Cleaned Output
print("\nFinal Preprocessed Data Head:")
print(df.head())
```

Output

1. Loading the Dataset

[1]:
```
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder, StandardScaler

# Load the dataset
df = pd.read_csv('streaming_platform_dataset.csv')

# Display initial shape and first few rows
print(f"Dataset Shape: {df.shape}")
df.head()
```

```
Dataset Shape: (515, 9)
```

[1]:

	user_id	age	subscription_tier	device_type	watch_duration_min	completion_pct	genre	country	monthly_spend
0	1329	58.0	Basic	Laptop	80.0	59.3	Action	Japan	199
1	1354	46.0	Free	Laptop	148.0	57.6	Horror	UK	0
2	1351	56.0	Premium	Tablet	177.0	48.5	Drama	USA	549
3	1151	19.0	Premium	Tablet	109.0	70.1	Documentary	USA	499
4	1279	16.0	Premium	Mobile	138.0	61.2	Drama	India	499

Original dataset loaded – Shape: (515, 9)

2. Checking Duplicates and Missing Values

[2]:
```
# Check total duplicate rows
print(f"Total Duplicate Rows: {df.duplicated().sum()}\n")

# Check missing values per column
print("Missing Values Per Column:")
print(df.isnull().sum())
```

```
Total Duplicate Rows: 13

Missing Values Per Column:
user_id               0
age                  40
subscription_tier     0
device_type           0
watch_duration_min    30
completion_pct        25
genre                 0
country               20
monthly_spend         0
dtype: int64
```

13 duplicate rows found; missing values present in age, watch_duration_min, completion_pct, and country

3. Removing Duplicate Records

[3]:
```
# Drop duplicate entries
df = df.drop_duplicates()

print(f"Shape after removing duplicates: {df.shape}")
```

```
Shape after removing duplicates: (502, 9)
```

Shape after removing duplicates: (502, 9)

4. Encoding Categorical Data

[5]:
```
# Convert categorical text columns to numerical representations
categorical_cols = ['subscription_tier', 'device_type', 'genre', 'country']
label_encoder = LabelEncoder()

for col in categorical_cols:
    df[col] = label_encoder.fit_transform(df[col])

# View transformed categorical columns
df[['subscription_tier', 'device_type', 'genre', 'country']].head()
```

[5]:

	subscription_tier	device_type	genre	country
0	4	0	0	5
1	6	0	4	6
2	8	3	3	7
3	8	3	2	7
4	8	1	3	4

subscription_tier, device_type, genre, and country encoded into numeric labels using LabelEncoder

5. Feature Scaling

```
[6]: # Standardize continuous numerical features (mean = 0, std = 1)
scaler = StandardScaler()
scale_features = ['age', 'watch_duration_min', 'completion_pct', 'monthly_spend']

df[scale_features] = scaler.fit_transform(df[scale_features])

# Verify scaling stats
df[scale_features].describe().round(2)
```

[6]:

	age	watch_duration_min	completion_pct	monthly_spend
count	502.00	502.00	502.00	502.00
mean	0.00	0.00	-0.00	0.00
std	1.00	1.00	1.00	1.00
min	-1.74	-0.18	-2.64	-1.17
25%	-0.84	-0.12	-0.68	-1.17
50%	0.00	-0.07	0.00	-0.20
75%	0.88	0.00	0.74	1.26
max	1.70	15.77	1.62	1.50

age, watch_duration_min, completion_pct, and monthly_spend standardized to mean = 0, std = 1

6. Final Preprocessed Dataset Info

```
[7]: # Display the cleaned and standardized dataset
print(f"Final Preprocessed Shape: {df.shape}\n")
print(df.info())
df.head()
```

```
Final Preprocessed Shape: (502, 9)

<class 'pandas.core.frame.DataFrame'>
Index: 502 entries, 0 to 514
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   user_id             502 non-null    int64
 1   age                 502 non-null    float64
 2   subscription_tier   502 non-null    int64
 3   device_type         502 non-null    int64
 4   watch_duration_min  502 non-null    float64
 5   completion_pct      502 non-null    float64
 6   genre               502 non-null    int64
 7   country             502 non-null    int64
 8   monthly_spend       502 non-null    float64
dtypes: float64(4), int64(5)
memory usage: 39.2 KB
None
```

Final shape: (502, 9) with no missing values across all columns

7. Final Preprocessed Dataset (Head)

[7]:

	user_id	age	subscription_tier	device_type	watch_duration_min	completion_pct	genre	country	monthly_spend
0	1329	1.221756	4	0	-0.046448	-0.110755	0	5	-0.199899
1	1354	0.395226	6	0	0.061957	-0.183166	4	6	-1.168445
2	1351	1.084001	8	3	0.108189	-0.570777	3	7	1.503575
3	1151	-1.464468	8	3	-0.000216	0.349267	2	7	1.260222
4	1279	-1.671100	8	1	0.046015	-0.029825	3	4	1.260222

First 5 rows of the cleaned, encoded, and scaled dataset

Result

Thus, the synthetic streaming platform dataset was successfully preprocessed using Python. 13 duplicate records were removed, missing values in age, watch_duration_min, completion_pct, and country were imputed using mean/mode, categorical columns (subscription_tier, device_type, genre, country) were label-encoded, and numerical

features (age, watch_duration_min, completion_pct, monthly_spend) were standardized. The final dataset (502 rows × 9 columns) is clean and ready for further analysis or machine learning.